JRC

JRC ERP — How it works

Jazallah Road Contracting · one system for estimating, site delivery, supply chain, billing, finance, people and plant

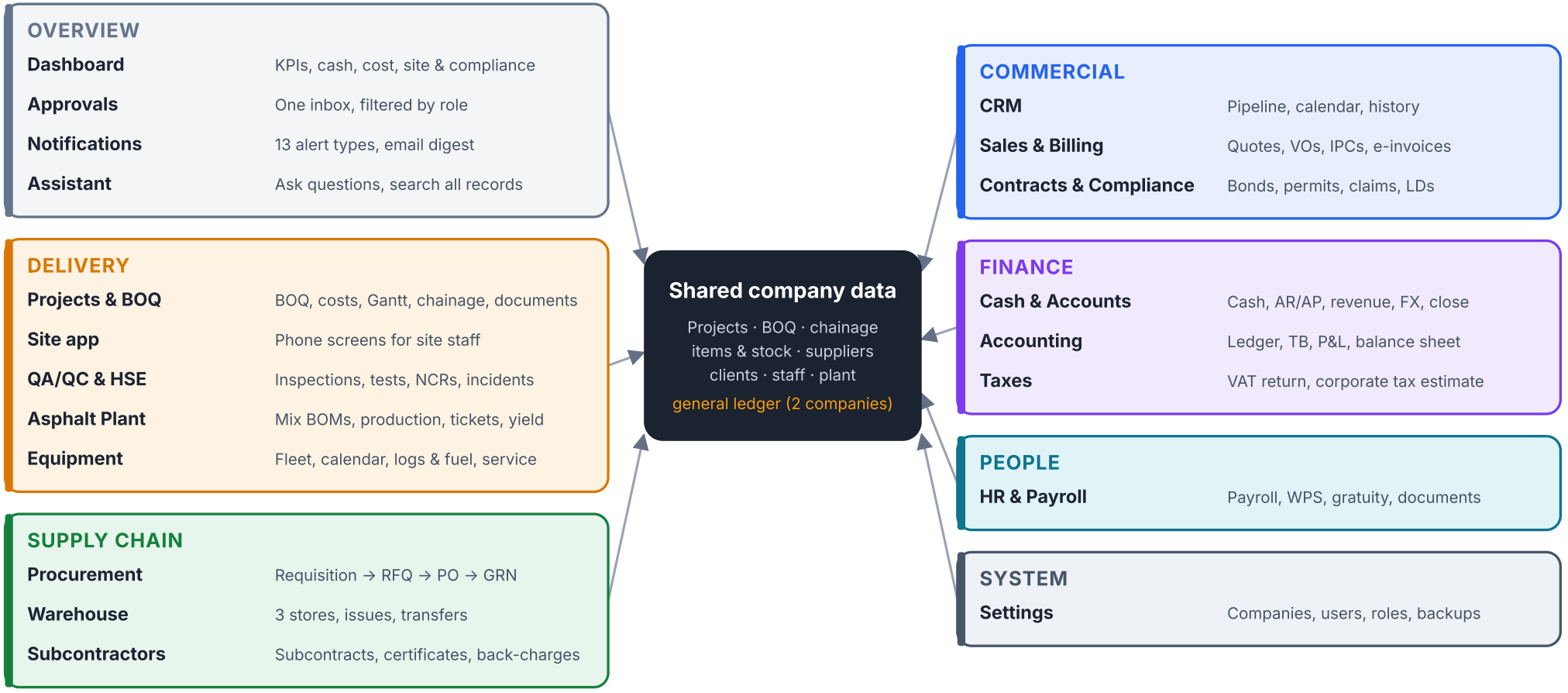
1 Module map
2 The project lifecycle
3 Win work → get paid
4 Buy materials → pay suppliers
5 Payment certificate maths
6 People & plant
7 Road operations
8 Accounting & group companies
9 Approvals, alerts & what updates what
10 Under the hood

All company details, people and figures in the system and in this guide are fictional samples. VAT and WPS outputs are illustrations to be checked with your accountant and bank.

1 · Module map

20 modules share one set of company data

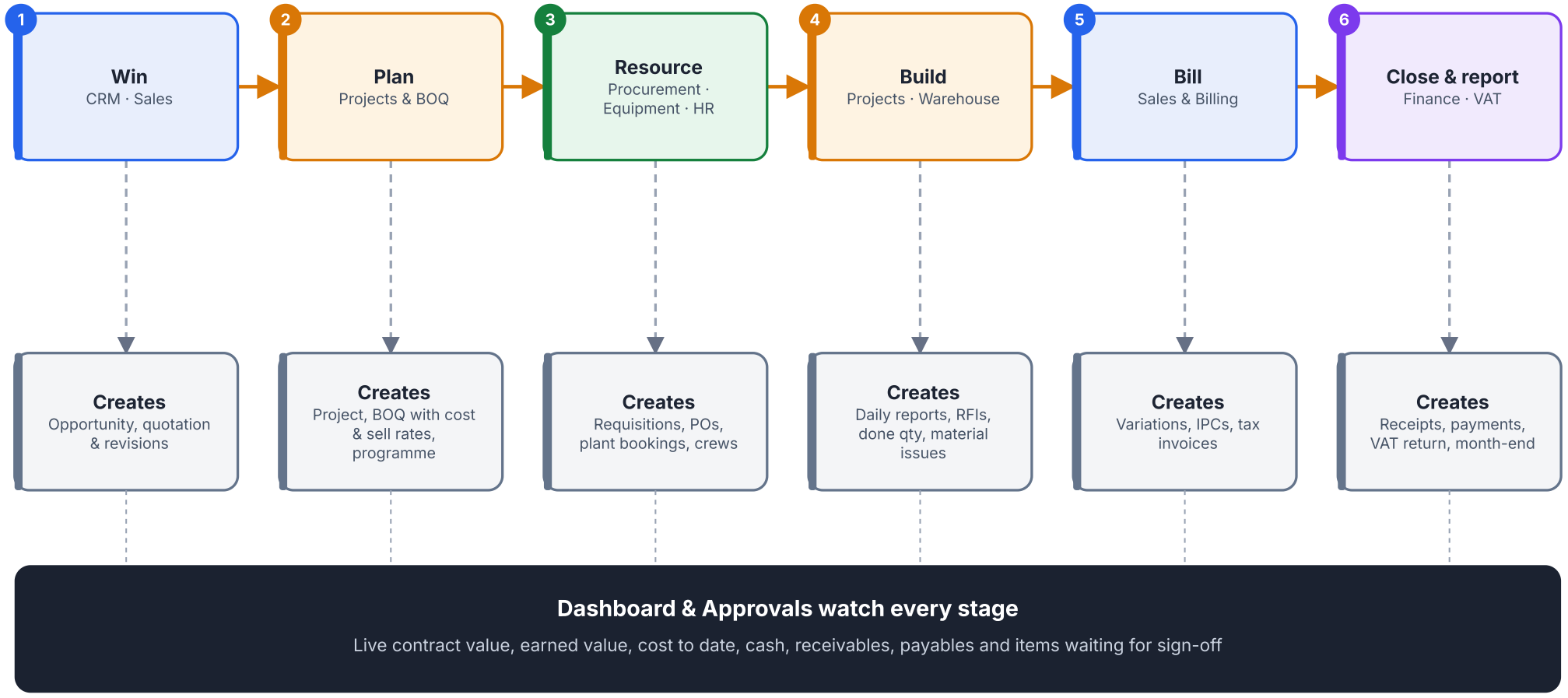
Every module reads from and writes to the same records: a quantity booked on site, a delivery received in the store, or an hour of overtime feeds straight into costs, invoices, VAT and the dashboard. Nothing is typed twice.



Delivery · Supply chain · Commercial · Finance · People · Overview · System

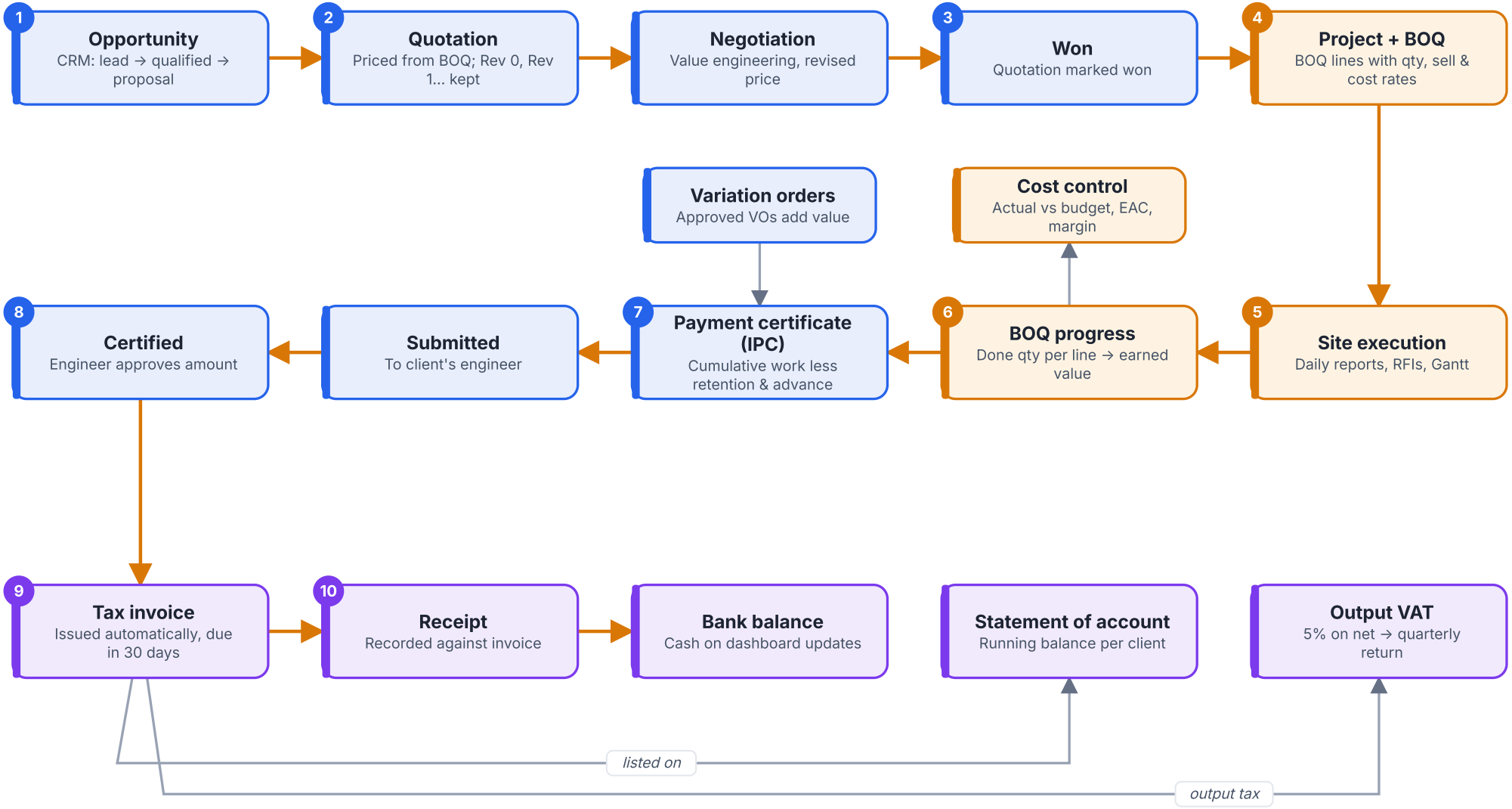
2 · The project lifecycle

A road project moves through six stages. Each stage is handled by one or more modules; the bottom row shows the records each stage creates.



3 · Win work → get paid

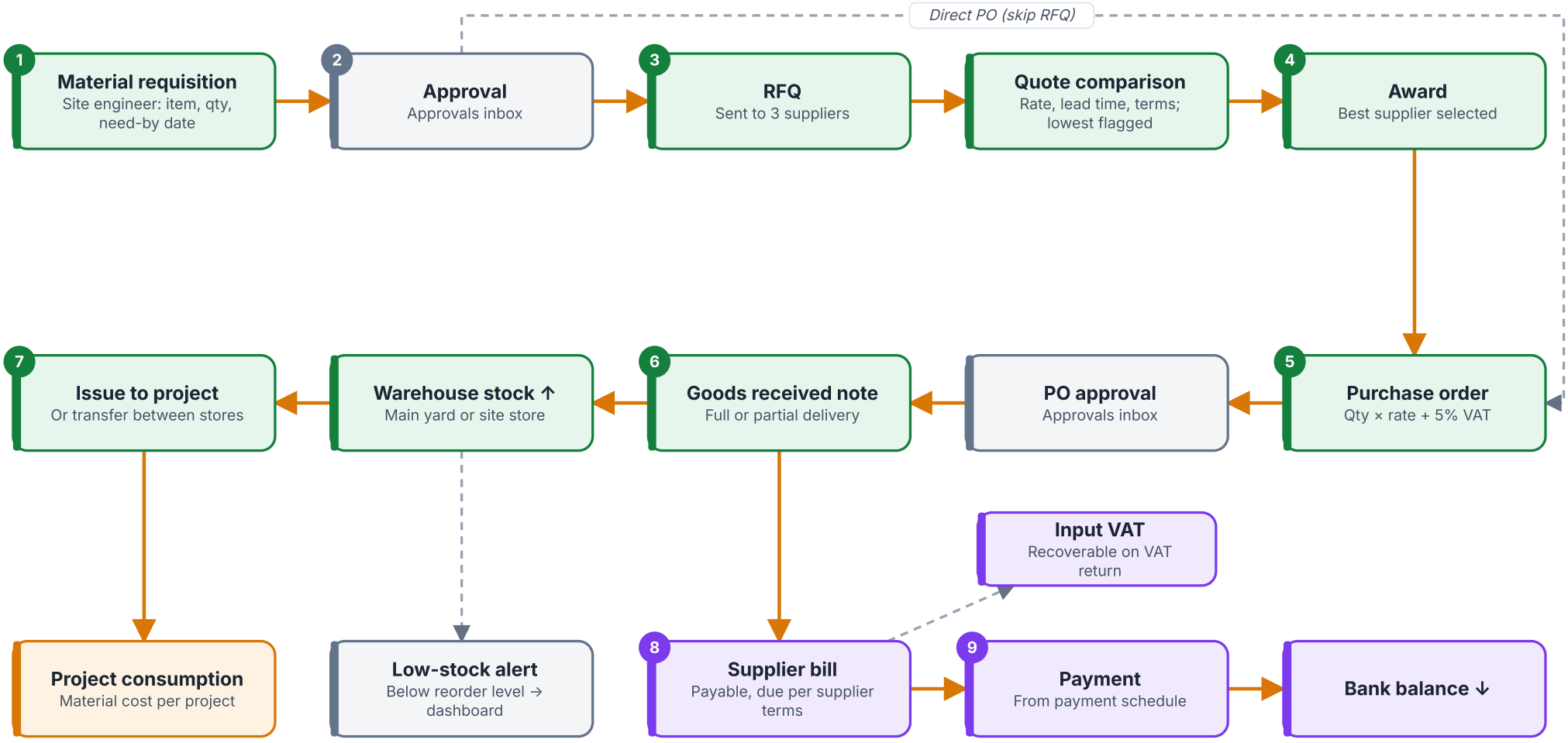
Follow the arrows: an opportunity becomes a priced quotation, then a project with a BOQ. Site progress against the BOQ produces the payment certificate (IPC), which becomes a tax invoice and finally cash in the bank.



JRC 4 · Buy materials → pay suppliers

Procurement · Warehouse · Payables · VAT

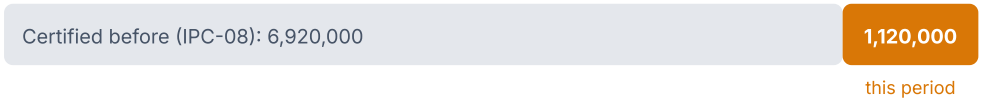
Site asks for material, the office gets competing quotes, the winner gets a PO, the store receives it, and the material is issued to the project where it becomes cost. The supplier's bill flows to payables and its VAT becomes recoverable input tax.



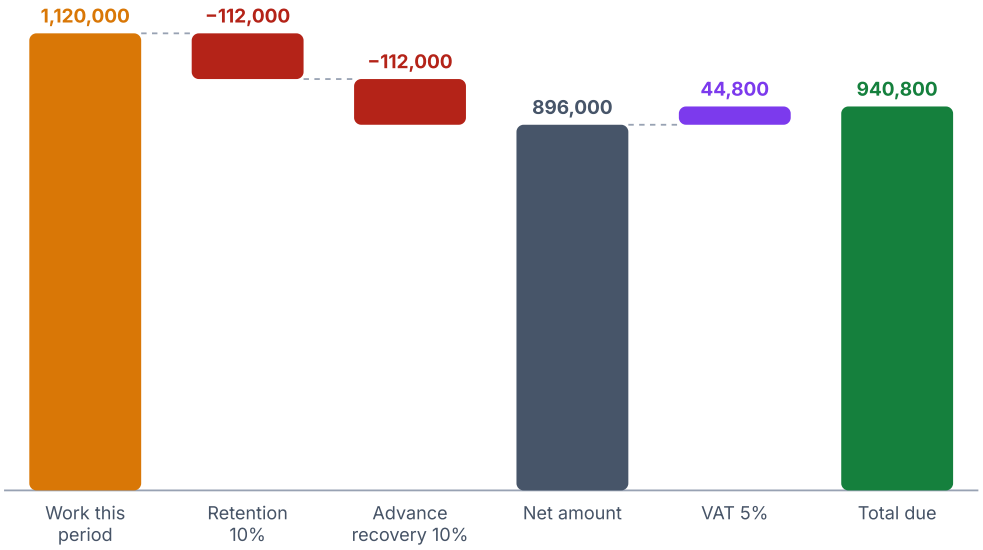
5 · Payment certificate (IPC) maths

Worked example: IPC-2401-09, Al Wathba Industrial Access Road

Cumulative work to date: AED 8,040,000



This certificate (AED)



Retention is held by the client (tracked under Receivables) and released at handover / end of defects liability.

Where cumulative work comes from

```
Earned value = Σ (BOQ done qty × sell rate)
Cumulative work = Earned value + approved VOs × % complete
```

- Site staff update **done quantity** per BOQ line.
- Approved **variation orders** are added pro-rata to progress.
- "New IPC from progress" takes this figure automatically.

Deductions (set in Settings)

```
Retention = 10% × work this period
Advance recovery = 10% × work this period
Net = work − retention − advance recovery
VAT = 5% × net · Total due = net + VAT
```

- Retention is tracked as money held by the client and released at handover / end of defects liability.

Status path

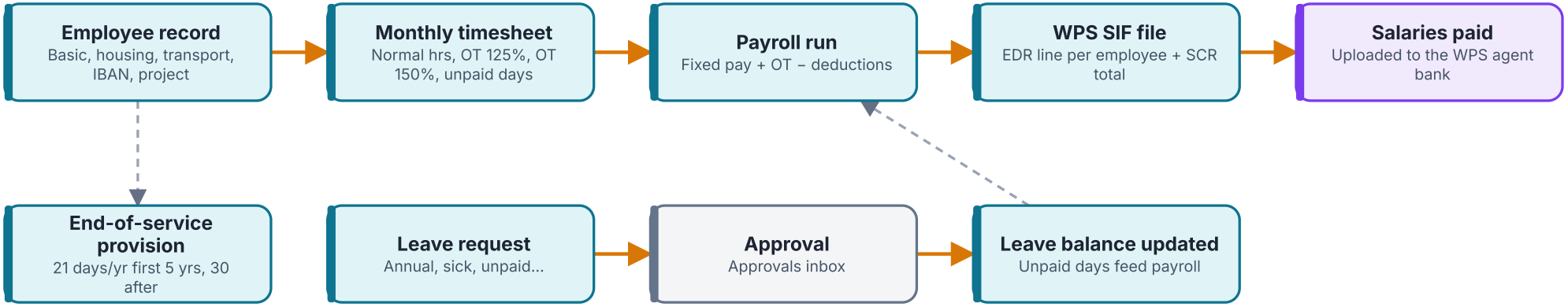
- **Draft** → **Submitted** to the engineer → **Certified** (a tax invoice is issued automatically) → **Paid** when the receipt clears the invoice.

Revenue recognition (Finance)

```
% complete = cost to date ÷ estimated total cost
Revenue to date = revised contract × % complete
Contract asset / (liability) = revenue − certified
```

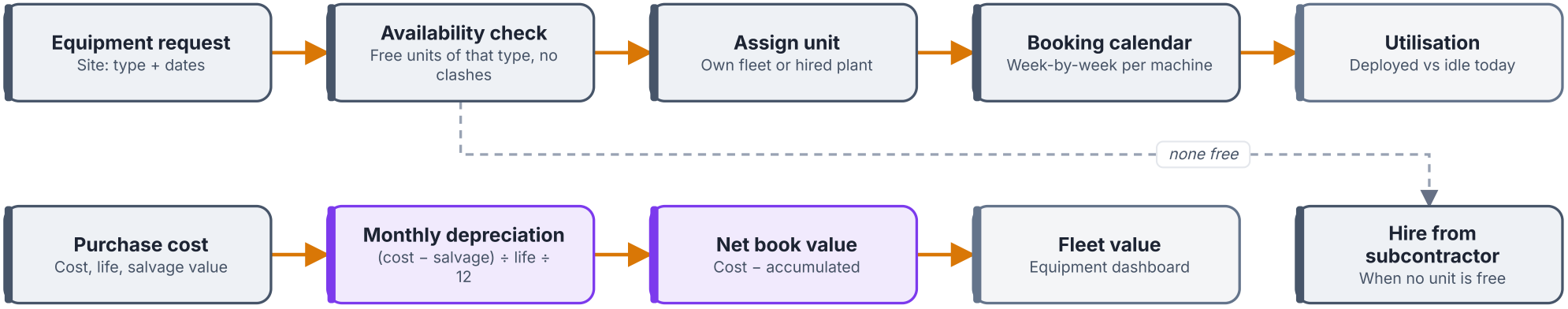
6 · People & plant

Hours worked become payroll and a bank-ready WPS file; service years drive the end-of-service provision. Equipment is requested by site, checked for clashes, booked on the calendar and depreciated monthly.



HR & PAYROLL ↑

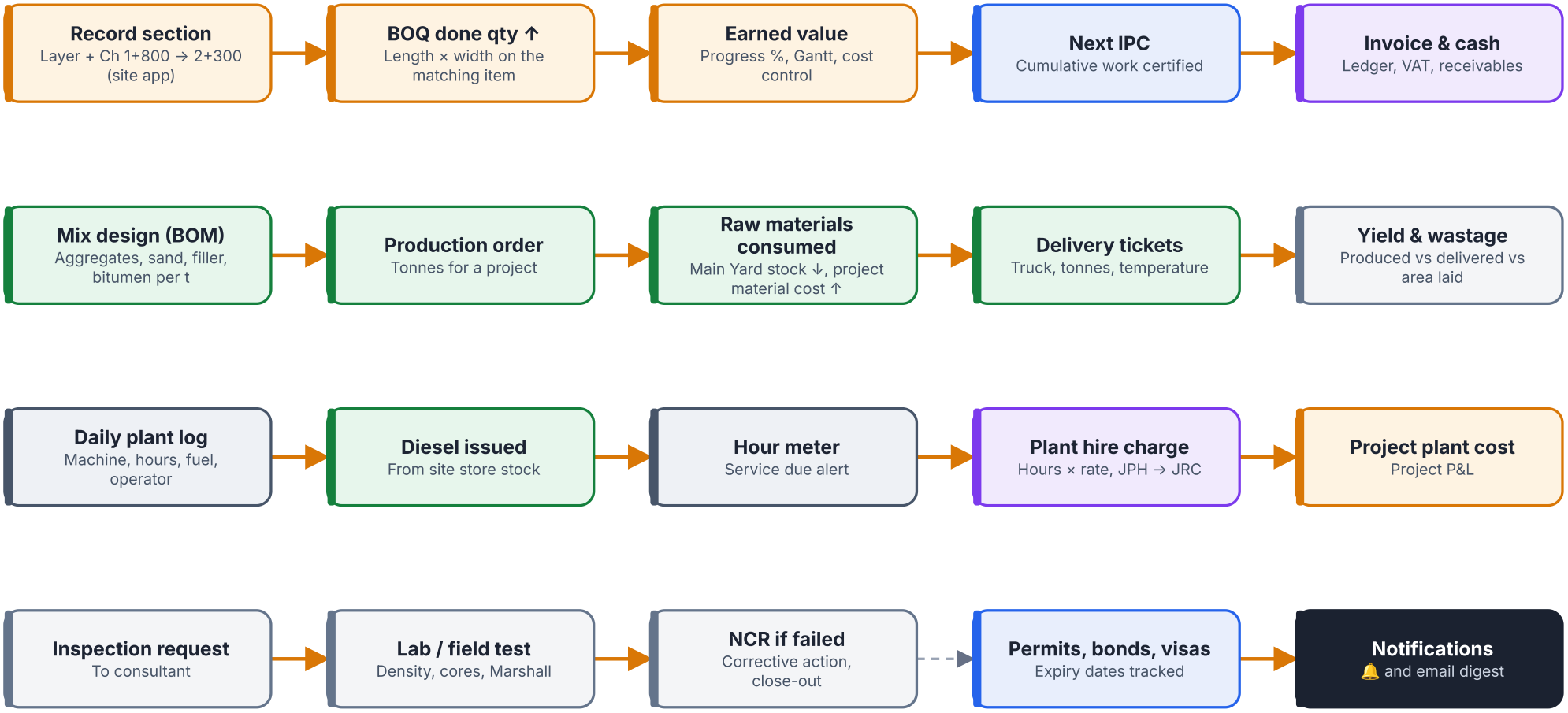
EQUIPMENT ↓



7 · Road operations

Chainage · Asphalt plant · Plant logs · QA/QC · Compliance

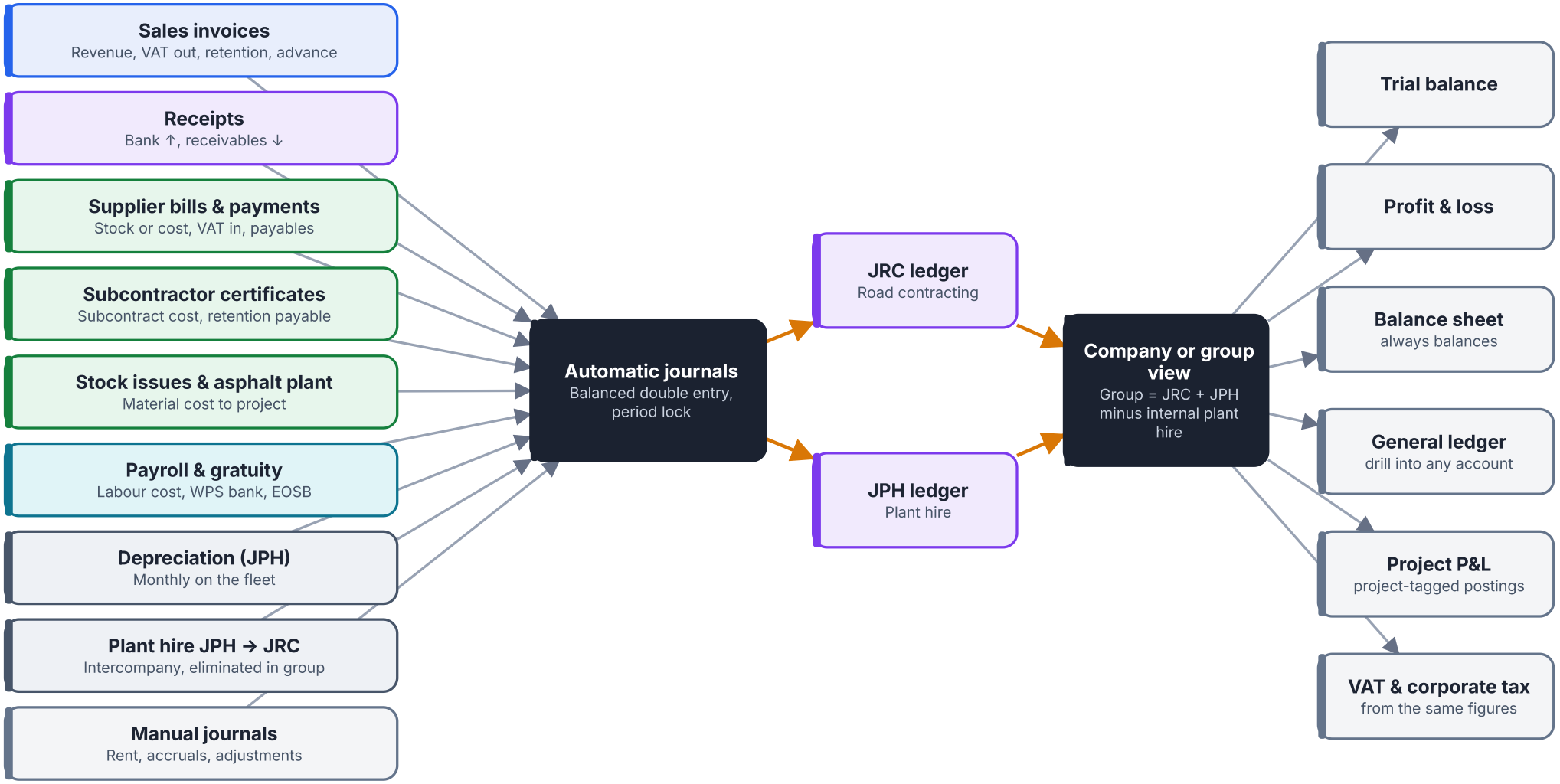
The road-specific modules feed the same records as everything else: a section recorded by chainage raises BOQ progress and the next payment certificate; asphalt production consumes stock; plant hours become project cost; failed tests and expiring permits raise alerts.



8 · Accounting & group companies

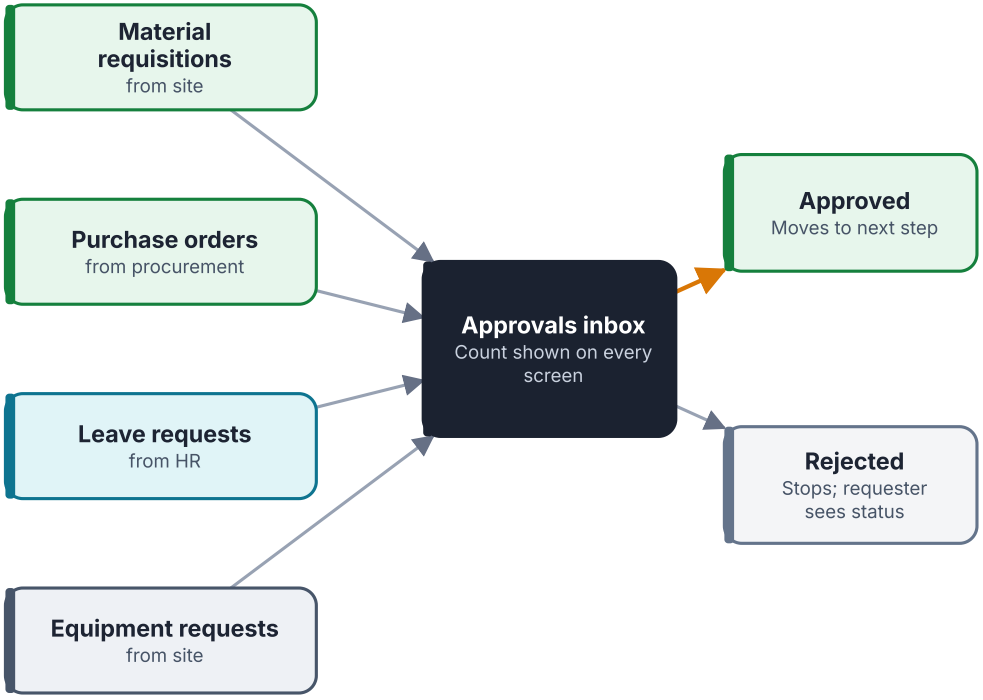
Every transaction posts itself to the ledger

Nobody keys journals for day-to-day work: invoices, receipts, bills, payroll, stock issues, depreciation and plant hire each create a balanced double entry. Two companies keep separate books — Jazallah Road Contracting and Jazallah Plant Hire (which owns the fleet and hires it to JRC) — and the group view removes the internal hire.



9 · Approvals, alerts & what updates what

One inbox · one action, many effects



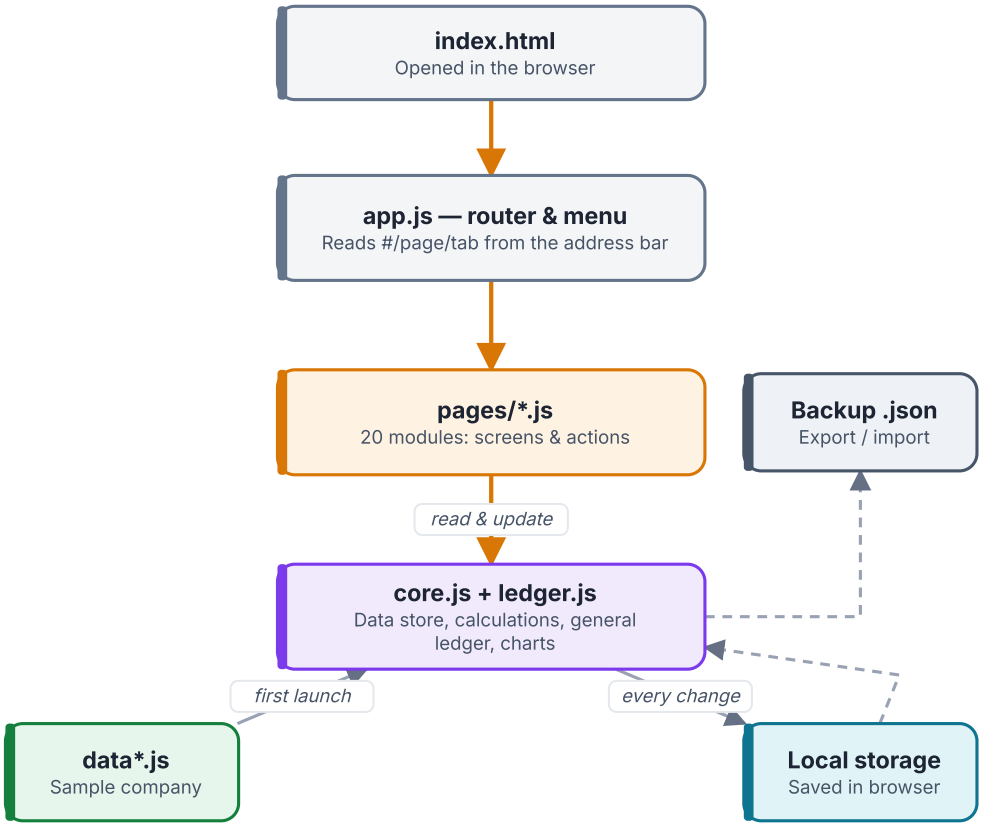
Approver per document and amount is set in Settings → Approval workflow

When you...	The ERP automatically...
Update BOQ done quantity	Recalculates earned value, progress %, Gantt bars, cost variance, forecast margin and the next IPC amount
Award an RFQ	Creates a PO (pending approval); adds a new supplier to the register if needed
Post a goods received note	Adds stock to the chosen warehouse, marks PO received, creates a supplier bill in payables
Issue stock to a project	Reduces warehouse stock, adds to project consumption, triggers low-stock alert if below reorder level
Certify an IPC	Issues a tax invoice due in 30 days; it appears in receivables, statement of account and output VAT
Record a receipt	Increases the bank balance, reduces receivables, marks the IPC paid when fully settled
Pay a supplier bill	Reduces the bank balance and clears the payable
Edit timesheet hours	Recalculates overtime pay, payroll total and the WPS file
Approve annual leave	Deducts days from the employee's leave balance
Assign equipment	Books it on the calendar (blocking clashes) and updates fleet utilisation
Approve a variation order	Raises the revised contract value and future IPC amounts
Record a chainage section	Adds length × width to the matching BOQ line, moves the schedule bar and earned value
Produce an asphalt order	Consumes aggregates, sand, filler and bitumen from Main Yard; posts material cost to the project
Log plant hours with fuel	Issues diesel from the site store, advances the hour meter (service alerts), charges plant hire to the project
Approve a subcontractor certificate	Creates a payable net of retention and back-charges; posts subcontract cost
Reject an inspection request	Optionally raises an NCR that stays open until closed out

Notifications (🔔)

- Overdue invoices and bills, low stock, items to approve
- Bonds, permits and staff documents nearing expiry
- Failed tests, open NCRs, plant service due, delay (LD) exposure, milestones due

10 · Under the hood



How it runs

- Plain HTML, CSS and JavaScript — no installation, no server, no build step.
- Open **erp/index.html** in any modern browser.
- Each screen is a page module; the address bar hash (e.g. **#/projects/P-2401/chainage**) decides which one shows.
- The Site app can be installed on a phone's home screen when the ERP is served over the web.

Where the data lives

- First launch loads the sample company from **data.js**.
- Every change is saved in the browser's local storage, so it survives a refresh.
- Settings → **Export backup** downloads everything as one JSON file; **Import** restores it; **Reset** returns to the sample.

Limits of the demo

- Users are switched from a menu, not real logins; roles limit what each person sees but are not enforced by a server.
- E-invoices are generated as XML files; email opens your mail app.
- Data is per browser; colleagues don't see each other's changes.

Path to production

- Move the data store to a hosted database (e.g. Supabase) with user accounts and role-based access.
- Keep the same screens and calculations — only the store layer changes.
- Add email notifications, a mobile layout for site staff and document attachments.